Classical Frequentist Statistics

MGTA 495: Marketing Analytics

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UCSD Rady | Spring 2026

Probability & Random Variables

Probability is about the uncertainty of outcomes $s \in S$ from the collection of all possible outcomes S

- For a coin flip, $s = \{H\}$ or $s = \{T\}$
- For tomorrow's weather, $s \in \{\text{sunny}, \text{rainy}, \text{cloudy}\}$
- For a customer's wait time at a store, $s \in [0, \infty)$ minutes

A random variable is a numeric mapping (or representation) of those outcomes $X : S \to \mathbb{R}$

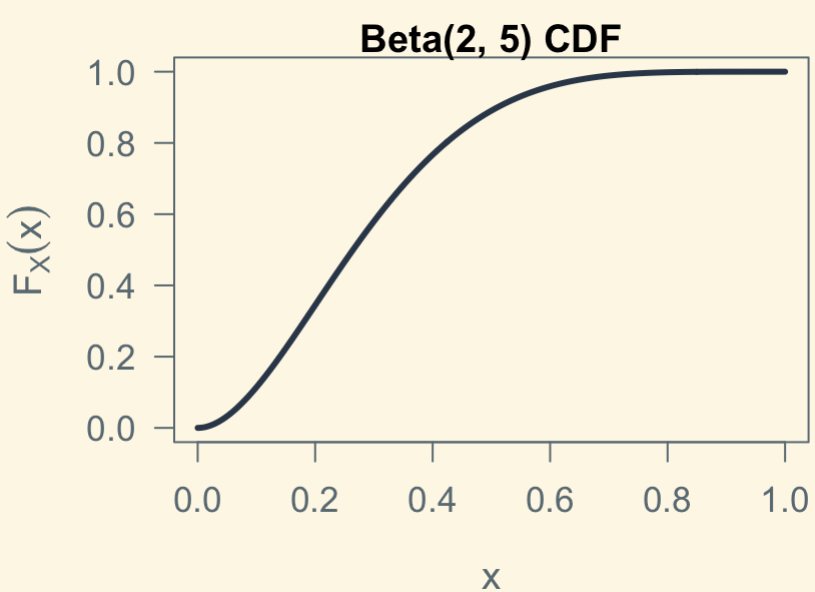
- $X(H) = 1$ and $X(T) = 0$
- $X(\text{sunny}) = 1,\ X(\text{rainy}) = 2,\ X(\text{cloudy}) = 3$ ← arbitrary choice
- $X(\text{wait time}) = \text{the wait time itself}$ ← natural mapping

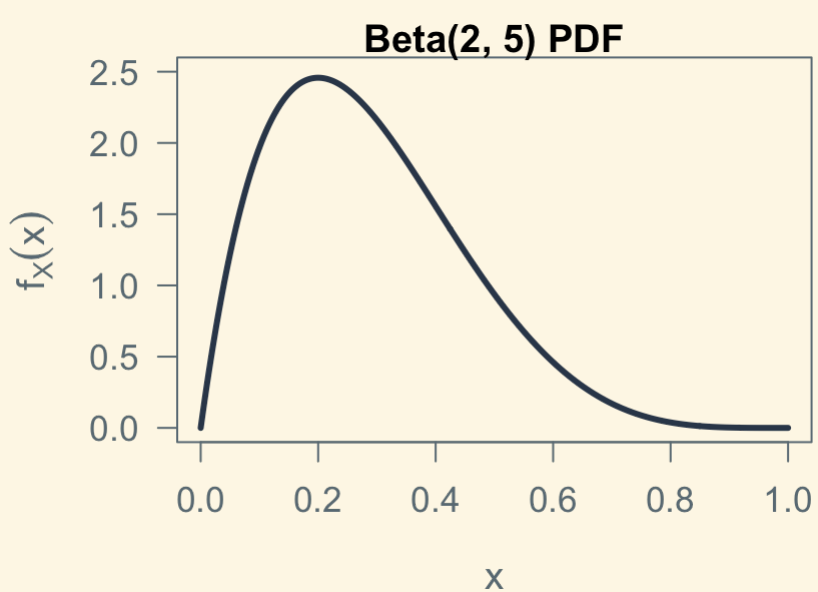
CDF and PDF

THE fundamental feature of X is its **cumulative distribution function** $F_X(x)$ and its derivative, the **probability density function** $f_X(x)$

$$\begin{aligned} F_X(x) &= P_X(X \leq x) \\ &= P_X\big(\,(-\infty, x)\,\big) \\ &= P_X\big(\,\{s \in S : X(s) \leq x\}\,\big) \end{aligned}$$

$$f_X(x) = F'_X(x)$$





What is Probability?

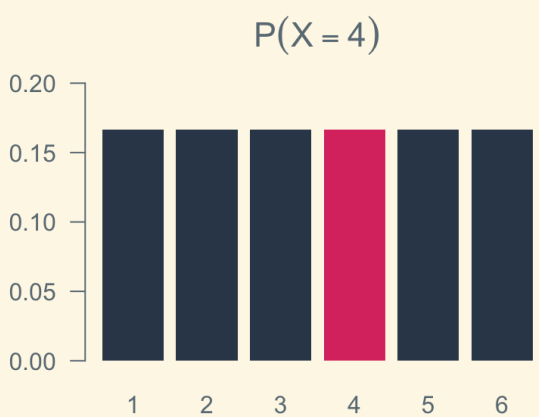
So far we've said what probability is *about* — but what *is* it?

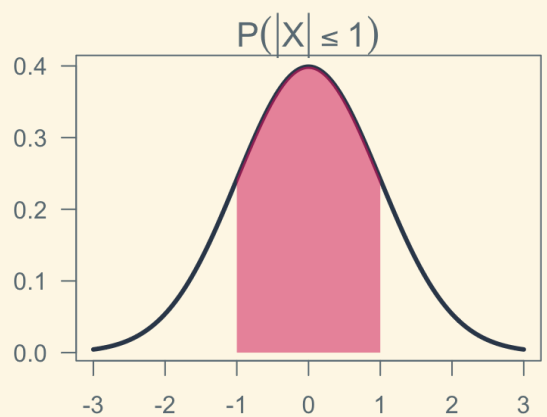
The **frequentist** answer: probability is the **long-run relative frequency** of an event over many repeated trials

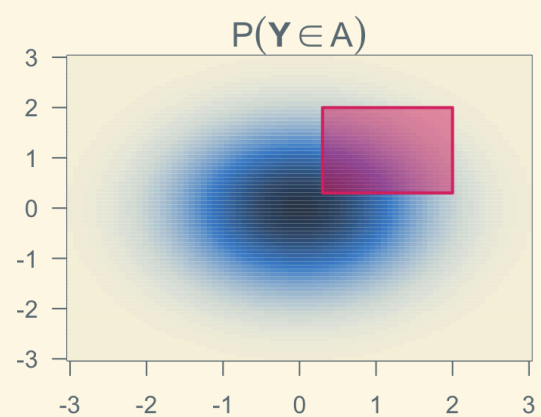
Discrete:
frequency of a *value*

Continuous:
frequency of a *range*

Multivariate:
frequency of a *region*





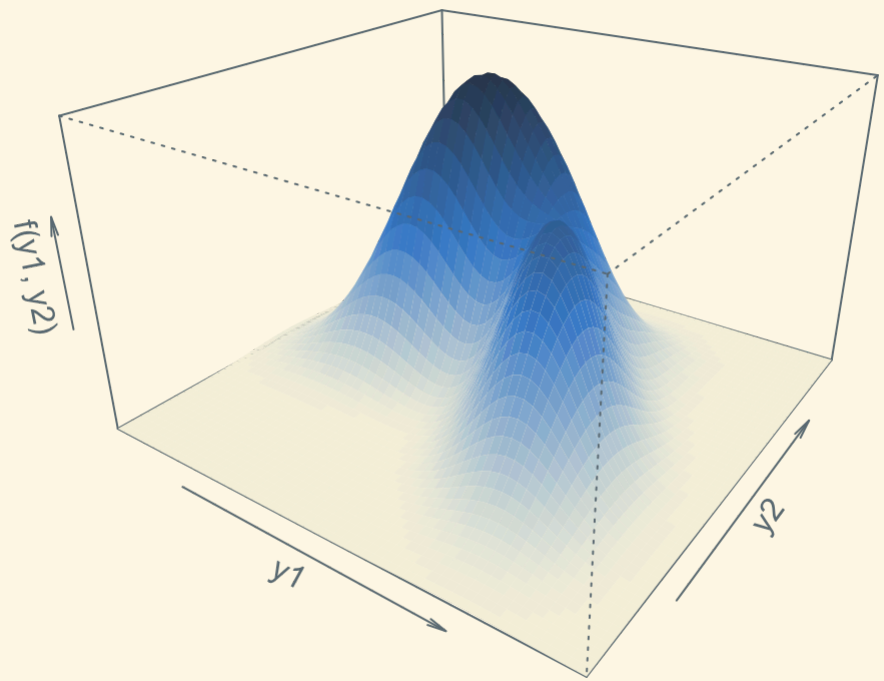


Vectors & Joint Density

Consider now p (not N or K) random variables that we group together in an n-dimensional **random vector** $\boldsymbol{Y} = (Y_1, \ldots, Y_p)$

$\boldsymbol{Y}$ has a **joint distribution** $F_{\boldsymbol{Y}}(\boldsymbol{y})$ with associated **joint density** $f_{\boldsymbol{Y}}(\boldsymbol{y})$

Example has $p = 2$: joint density $f_{\boldsymbol{Y}}(y_1, y_2)$



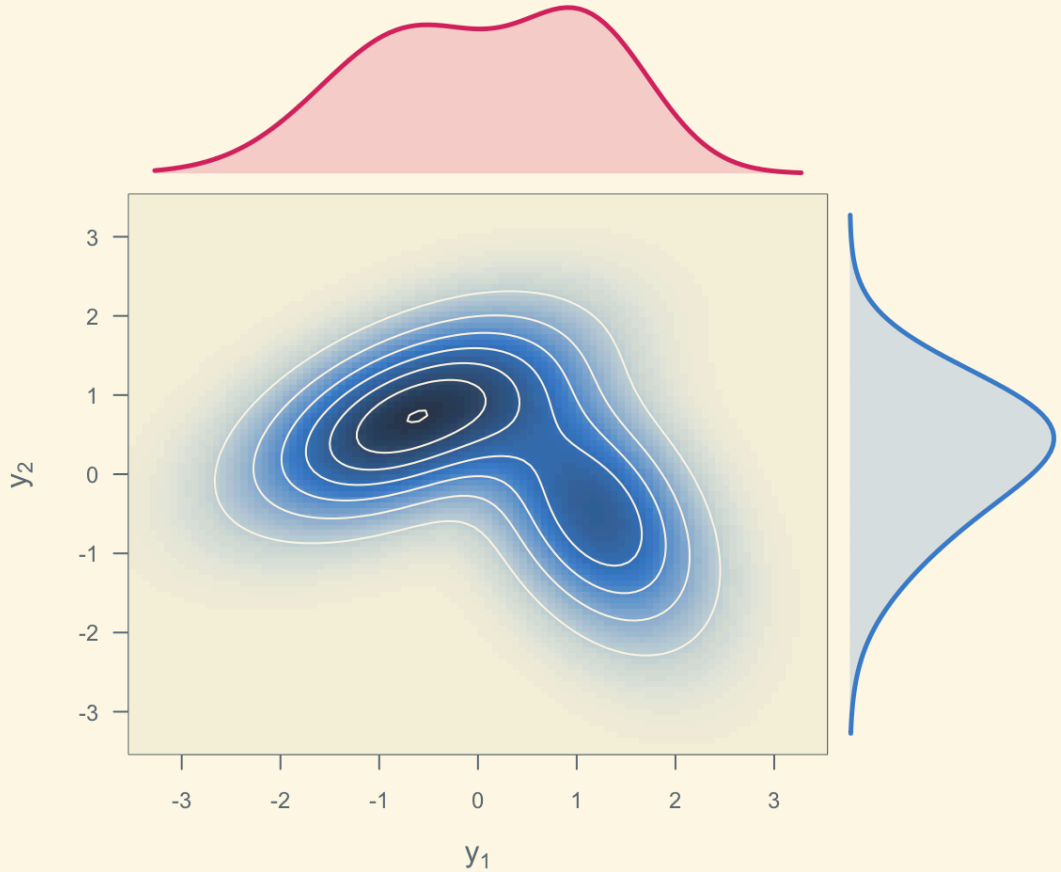
Marginal Distributions

The **marginal distribution** of Y_1 is obtained by integrating the joint over Y_2:

$$f_{Y_1}(y_1) = \int f_{Y_1,Y_2}(y_1, y_2)\, dy_2$$

Intuitions:

- **Stack:** collapse the 3D density onto one axis — pile up all the probability mass
- **Shadow:** the silhouette the density surface casts
- **Ignore:** the distribution of Y_1 when you don't know (or don't care about) Y_2



Conditional Distributions

The **conditional distribution** of Y_2 given $Y_1 = a$ is a **slice** through the joint density at $Y_1 = a$, rescaled to integrate to 1:

$$f_{Y_2|Y_1}(y_2 \mid y_1 = a) = \frac{f_{Y_1,Y_2}(a,\ y_2)}{f_{Y_1}(a)}$$

Row Vectors

In practice, we often have in mind a **row** vector of random variables where

- We label one of these as Y
- And label the maining K as $\boldsymbol{X} = (X_1, X_2, \ldots, X_K)$

$$(Y,\ X_1,\ X_2,\ \ldots,\ X_K) \qquad \text{with joint density } f_{Y,\boldsymbol{X}}(y, \boldsymbol{x})$$

We do this because the joint density $f_{Y,\boldsymbol{X}}(y, \boldsymbol{x})$ may be too hard to estimate – or simply not what we care about – but we can factor

$$f_{Y,\boldsymbol{X}}(y, \boldsymbol{x}) = f_{Y|\boldsymbol{X}}(y|\boldsymbol{x}) \cdot f_{\boldsymbol{X}}(\boldsymbol{x})$$

and we might focus or model only $f_{Y|\boldsymbol{X}}$, ignoring $f_{\boldsymbol{X}}$
(this is what **regression** is all about!)

Column Vectors

Alternatively, we might have in mind the idea of measuring something repeatedly, giving rise to a **column** vector of random variables:

$$\boldsymbol{Y} = (Y_1, Y_2, \ldots, Y_n)$$

In **cross sectional analysis**, we often make the **iid assumption** that each Y_i is **independent** of all other Y's and each follows an **identical** marginal distribution

$$f_{\mathbf{Y}}(\boldsymbol{y}) = \prod_{i=1}^{n} f_Y(y_i)$$

In **time series analysis**, we replace n with T and factor the joint via the chain rule into a **product of conditionals** — each depends on the past:

A Dataset

A **dataset** is a matrix of random variables

Stack N row vectors $(Y_i, X_{1i}, \ldots, X_{Ki})$ each of length $K = 1$

$$\boldsymbol{D} = \begin{bmatrix} Y_1 & X_{11} & X_{21} & \cdots & X_{K1} \\ Y_2 & X_{12} & X_{22} & \cdots & X_{K2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ Y_N & X_{1N} & X_{2N} & \cdots & X_{KN} \end{bmatrix}$$

Every entry is a random variable, every row (and column) is a random vector, and the entire matrix has its own joint distribution of dimension $n(K+1)$ —- *enormous*

Nothing has been observed yet —- this (along with any assumptions about its structure) is the **data-generating process**

The Sample

The **sample** is what we observe – a **realization** of $\boldsymbol{D}$:

$$\boldsymbol{d} = \begin{bmatrix} y_1 & x_{11} & x_{21} & \cdots & x_{K1} \\ y_2 & x_{12} & x_{22} & \cdots & x_{K2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ y_n & x_{1n} & x_{2n} & \cdots & x_{Kn} \end{bmatrix}$$

Two ways to think about a sample:

1. as **one draw** from the enormous joint distribution $f_{\boldsymbol{D}}(\boldsymbol{d})$
2. as N **draws** of $f_{Y|\boldsymbol{X}}(y|\boldsymbol{x})$ stacked together – implicitly assuming at least independence across rows

Assumptions

Learning about the full joint distribution of $\boldsymbol{D}$ from a single realization is hopeless — too many dimensions, too little data

Progress requires **assumptions** that shrink the problem:

1. **Focus on $f_{Y|\boldsymbol{X}}$ rather than the full joint** — we only need to model how Y relates to $\boldsymbol{X}$, not the distribution of $\boldsymbol{X}$ itself
2. **Rows are iid** — each observation contributes its own information, so the joint column-wise conditional distribution factors into a product of conditionals:

$$f_{\boldsymbol{Y}|\boldsymbol{X}_1,\ldots,\boldsymbol{X}_K}(\boldsymbol{y}|\boldsymbol{x_1},\ldots,\boldsymbol{x}_K) = \prod_{i=1}^{N} f_{Y_i|\boldsymbol{X}_i}(y_i \mid \boldsymbol{x}_i)$$

pay careful attention to what is a row-vector vs col-vector in the notation

so N observations accumulate evidence about one target: $f_{Y|\boldsymbol{X}}(Y|\boldsymbol{x})$

Parametric Family of Distributions

$f_{Y|\boldsymbol{X}}(y \mid \boldsymbol{x})$ may still be too general – Often it is reasonable to assume it belongs to some **family** of distributions $\mathcal{F}$

Specifically, assume $\mathcal{F}$ is a **parametric family** $\mathcal{F}_\theta$ — the "type" of distribution is known up to some unknown **parameter(s)** $\boldsymbol{\theta} = \{\theta_1, \theta_2, \ldots, \theta_J\}$

Combined with assumptions from the previous slide, this gives:

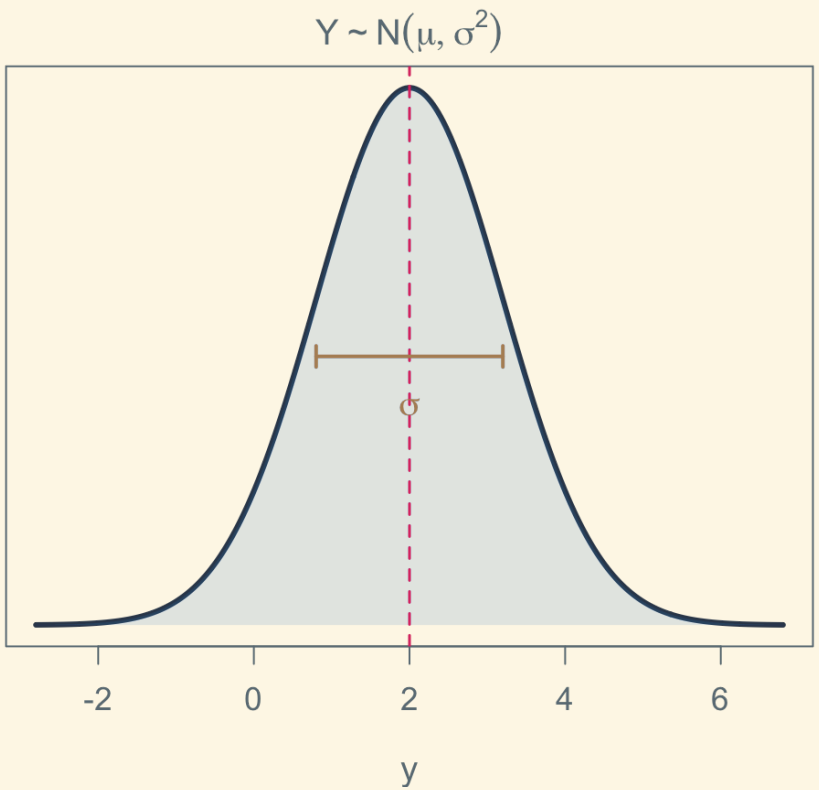
$$f_{Y|\boldsymbol{X}}(y \mid \boldsymbol{x}, \boldsymbol{\theta}) = f_{\boldsymbol{\theta}}(y \mid \boldsymbol{x}) \in \mathcal{F}_{\boldsymbol{\theta}}$$

The entire data-generating process is now characterized by the small set of parameters $\boldsymbol{\theta}$ — and the statistical task becomes: *learn $\boldsymbol{\theta}$ from the data* rather than *learn $f_{Y|\boldsymbol{X}}$ from the data*

Example: Gaussian ("Normal"), No X

Suppose $Y_i \sim \mathcal{N}(\mu, \sigma^2)$, then:

- $\theta = (\mu, \sigma)$
- $f_{\mu,\sigma}(y) = (2\pi\sigma^2)^{-1/2}\, e^{-\frac{1}{2}\left(\frac{y-\mu}{\sigma}\right)^2}$



Suppose also that the N Y_i are independent, then:

$$f_{\mathbf{Y}}(\boldsymbol{y}) = \prod_{i=1}^{N} f_{\mu,\sigma}(y_i)$$

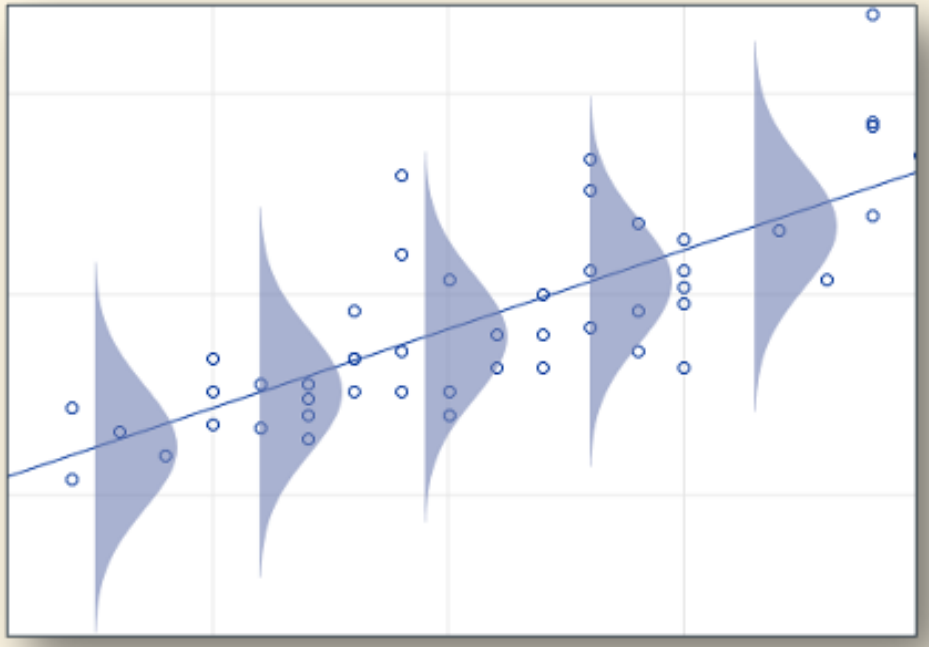
$$= (2\pi\sigma^2)^{-N/2}\, e^{-\frac{1}{2}\sum_{i=1}^{N}\left(\frac{y_i-\mu}{\sigma}\right)^2}$$

Example: Normal Linear Regression

A **regression function** $g(\boldsymbol{X})$ is a function of $\boldsymbol{X}$ used to predict Y

The "Classic" **Normal Linear Regression Model** uses $g(\boldsymbol{X}) = \boldsymbol{X}'\boldsymbol{\beta}$

$$Y_i = \beta_0 + \beta_1 X_1 + \cdots + \beta_K X_K + \varepsilon_i \qquad \varepsilon_i \sim N(0, \sigma^2)$$



Notice the model can be written as $Y_i | \boldsymbol{X}_i \sim N(\boldsymbol{X}_i' \boldsymbol{\beta}, \sigma^2)$

Example: Normal Linear Regression

$Y_i|\boldsymbol{X}_i \sim N(\boldsymbol{X}_i'\boldsymbol{\beta}, \sigma^2)$ means $f_{Y|\boldsymbol{X}}(y|\boldsymbol{x}, \boldsymbol{\beta}, \sigma) = (2\pi\sigma^2)^{-1/2}\, e^{-\frac{1}{2}\left(\frac{y-\boldsymbol{x}'\boldsymbol{\beta}}{\sigma}\right)^2}$

For the full dataset

$$
\begin{aligned}
f_{Y|\boldsymbol{X}}(y \mid \boldsymbol{x}, \boldsymbol{\beta}, \sigma) &= \prod_{i=1}^{n} f_{Y|\boldsymbol{X}}(y|\boldsymbol{x}, \boldsymbol{\beta}, \sigma) \\
&= (2\pi\sigma^2)^{-n/2}\, e^{-\frac{1}{2}\sum_{i=1}^{n}\left(\frac{y_i-\boldsymbol{x}_i'\boldsymbol{\beta}}{\sigma}\right)^2}
\end{aligned}
$$

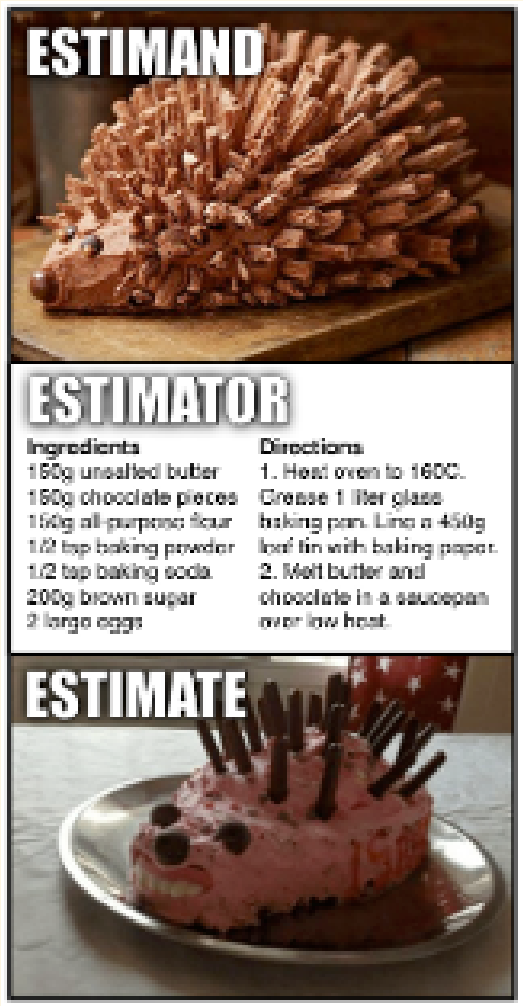
Statistics as Estimators

A **statistic** $t(\boldsymbol{Y})$ is any real or vector-valued function that can be computed using the data alone[1] (ie, an "algorithm")

A (point) **estimator** $\hat{\theta}(\boldsymbol{Y})$ of an unknown parameter (vector) θ – our **estimand** – is any statistic used for estimating θ

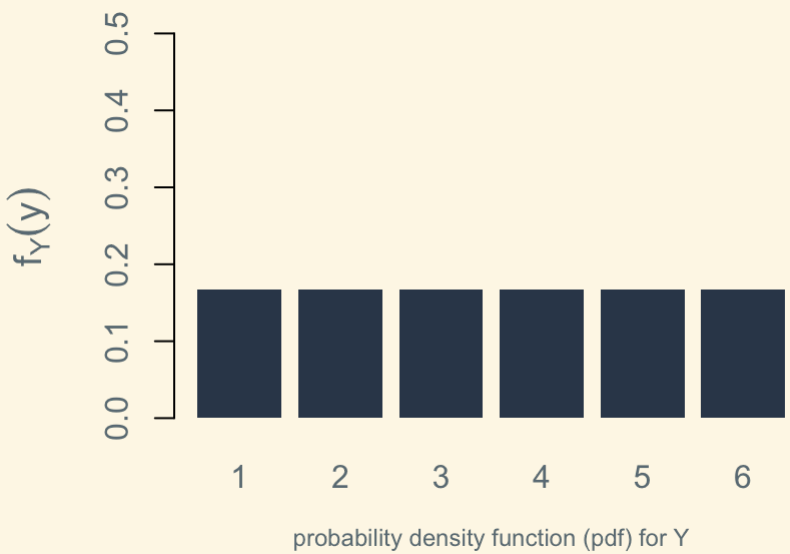
The value of $\hat{\theta}(\boldsymbol{y})$ evaluated for a given observed sample $\boldsymbol{y}$ is called an **estimate**

$\hat{\theta}$ is a function of a random sample, so it is a random variable, and thus it has its own distribution called the **sampling distribution**



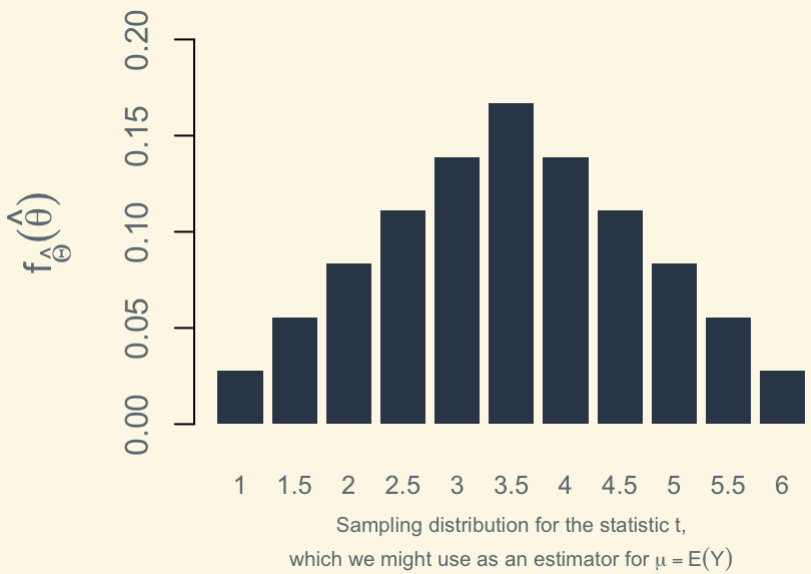
Example: Dice Sampling Distribution

Let Y be a roll of a die



Let the statistic

$$t(Y_1, Y_2) = (Y_1 + Y_2)/2$$



Where do Estimators come from?

There is no single recipe — different **estimation principles** lead to different estimators with different properties

- **Method of Moments** — equate population moments $\mathbb{E}[Y^k]$ with sample moments $\frac{1}{N}\sum y_i^k$ and solve for $\boldsymbol{\theta}$; simple but can be inefficient
- **Least Squares** — choose $\boldsymbol{\theta}$ to minimize $\sum_{i=1}^N (y_i - g(\boldsymbol{x}_i, \boldsymbol{\theta}))^2$; does not require a full distributional assumption
- **Maximum Likelihood** — choose $\boldsymbol{\theta}$ to maximize $\prod_{i=1}^N f_{\boldsymbol{\theta}}(y_i|\boldsymbol{x}_i)$; uses the full model; optimal (efficient) under correct specification
- **Bayesian** — combine a prior $\pi(\boldsymbol{\theta})$ with the likelihood via Bayes' rule to get a posterior $\pi(\boldsymbol{\theta}|\boldsymbol{y})$; the "estimator" is the entire posterior distribution
- **M-estimators** — generalize MLE: choose $\boldsymbol{\theta}$ to minimize $\sum_{i=1}^N \rho(y_i, \boldsymbol{\theta})$ for some loss ρ; includes least squares, MLE, quantile regression, and more as special cases

Properties of Estimators

Our general definition of $\hat{\theta}(\boldsymbol{Y})$ does not say anything about the "goodness" of the estimator; we are interested in "good" estimators

A somewhat naïve attempt to measure the goodness-of-estimation of $\hat{\theta}$ would be to consider the absolute error $|\hat{\theta} - \theta|$, but:

- $\hat{\theta}$ is a function of a random sample, so $|\hat{\theta} - \theta|$ will vary from sample to sample
- Convention is to use squares rather than absolute values

Thus, the better measure is an average squared difference over all possible samples, called the **mean squared error** (MSE) given by

$$MSE(\hat{\theta}, \theta) = \mathbb{E}\left[(\hat{\theta} - \theta)^2\right] \qquad = \int (\hat{\theta} - \theta)^2 f_{\hat{\theta}}(\hat{\theta})\, d\hat{\theta}$$

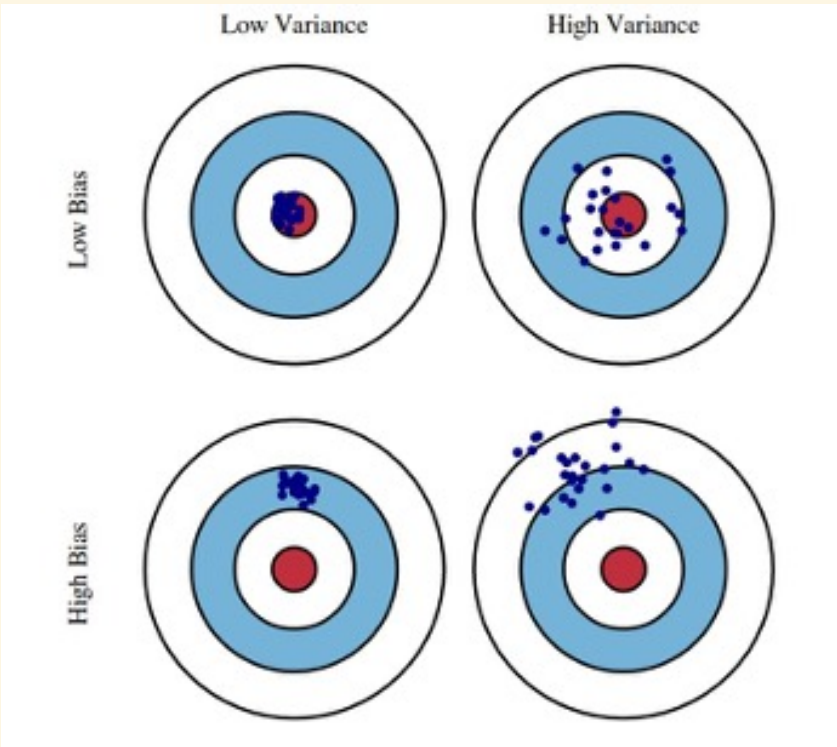
MSE Decomposition

The MSE can be decomposed into two components:

$$MSE(\hat{\theta}, \theta) = Var(\hat{\theta}) + Bias^2$$

- the **variance** is the stochastic error resulting from sampling variability
- the **bias** is systematic or deterministic error

Ideally, a good estimator (ie, small MSE) should have both low variance and low bias – **this is rare** because there is a **bias-variance tradeoff**; lowering one often increases the other



Unbiasedness

An estimator is **unbiased** if $\mathbb{E}[\hat{\theta}] = \theta$ — on average across repeated samples, it hits the true value

The *classical* approach is to restrict attention to unbiased estimators and then find the one with the smallest variance among them

Modern methods (regularization, shrinkage, machine learning) deliberately accept a small amount of bias in exchange for a large reduction in variance — often achieving lower MSE overall

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another day another reminder that economists' unrelenting focus on the unbiasedness property of point estimates has permantly warped our intuitive understanding that the usefuless and overall quality of any estimation exercise obviously involves the variance of the estimator

Standard Error

All estimators have error: if θ is estimated by $\hat{\theta}$, then $\hat{\theta} - \theta$ is different from 0

The standard measure of error is the MSE, and when $\hat{\theta}$ and the MSE are taken together, we get a feeling of how precise the estimate is

It is more common to quote the **standard error** $SE(\hat{\theta}) = \sqrt{MSE}$ because it's in the same units as $\hat{\theta}$

Note that for **unbiased** estimators, $MSE(\hat{\theta}, \theta) = Var(\hat{\theta})$ and thus
$SE(\hat{\theta}) = \sqrt{Var(\hat{\theta})} = sdev(\hat{\theta})$,
ie the standard deviation of the sampling distribution

LLN

Many parameters are a form of expectation

- e.g., if $Y \sim N(\mu, \sigma^2)$ then $\mu = E[Y]$ and $\sigma^2 = E[(Y - \mu)^2]$
- e.g., if $Y \sim Bernoulli(p)$ then $p = E[Y]$

The **Law of Large Numbers (LLN)** says that averages converge to expectations as n gets large:

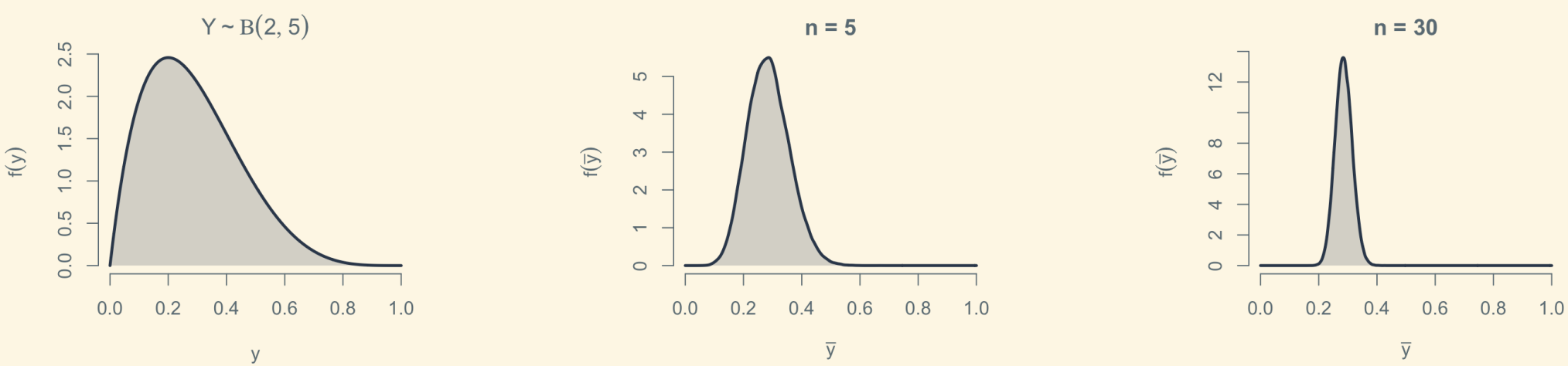
$$\bar{Y} = \frac{1}{n} \sum_{i=1}^{n} Y_i \qquad\qquad \bar{Y} \xrightarrow{n \to \infty} E[Y]$$

This makes $\bar{Y}$ a natural — and **consistent** — estimator of $E[Y]$, and many estimators are built on this principle

CLT

The **Central Limit Theorem (CLT)** says that the sampling distribution of an average converges to a Normal distribution as n gets large, regardless of the distribution of underlying variable:

$$Y_i \sim f(y;\theta) \qquad\qquad \bar{Y} \approx N(\mu_Y, \sigma_Y^2/n)$$



Normality

If θ is an expectation and we want to estimate it, we very well might use an estimator $\hat{\theta}$ that is some sort of average

- By LLN: $\hat{\theta}$ will be close to θ if n is large
- By CLT: the sampling distribution of $\hat{\theta}$ will be approximately normal

Confidence Interval

A $(1-\alpha) \times 100\%$ **confidence interval** for θ is the pair of scalar valued statistics $L = L(Y_1, \ldots, Y_n)$ and $U = U(Y_1, \ldots, Y_n)$ such that $\Pr(L \leq \theta \leq U) = 1 - \alpha$

When the sampling distribution of $\hat{\theta}$ is (approximately) normal, a 95% confidence interval (ie $\alpha = 0.05$) is constructed using

- $L = \hat{\theta} - 1.96 * SE(\hat{\theta})$
- $U = \hat{\theta} + 1.96 * SE(\hat{\theta})$

No statistical theory dictate a "correct" choice for the **significance level** α, but 0.05 has traditionally been most common

Note that, in general, confidence intervals are not necessarily unique and not necessarily symmetrical

Bootstrap

To construct a confidence interval we need $SE(\hat{\theta})$, the standard deviation of the sampling distribution of $\hat{\theta}$.

But we only have one sample — we can't actually repeat the experiment thousands of times

The **bootstrap** solves this by *resampling from the data we have* to approximate the sampling distribution

Algorithm:

1. From your observed sample of size n, draw n observations with replacement — call this a bootstrap sample
2. Compute $\hat{\theta}^*$ on the bootstrap sample
3. Repeat steps 1–2 a total of B times (eg $B = 10{,}000$)
4. The standard deviation of $\hat{\theta}_1^*, \ldots, \hat{\theta}_B^*$ is your bootstrap SE

Interpreting Confidence Intervals

Suppose we estimate μ with $\bar{Y}$ and we find $\bar{Y} = 10$ and $SE = 0.2$.

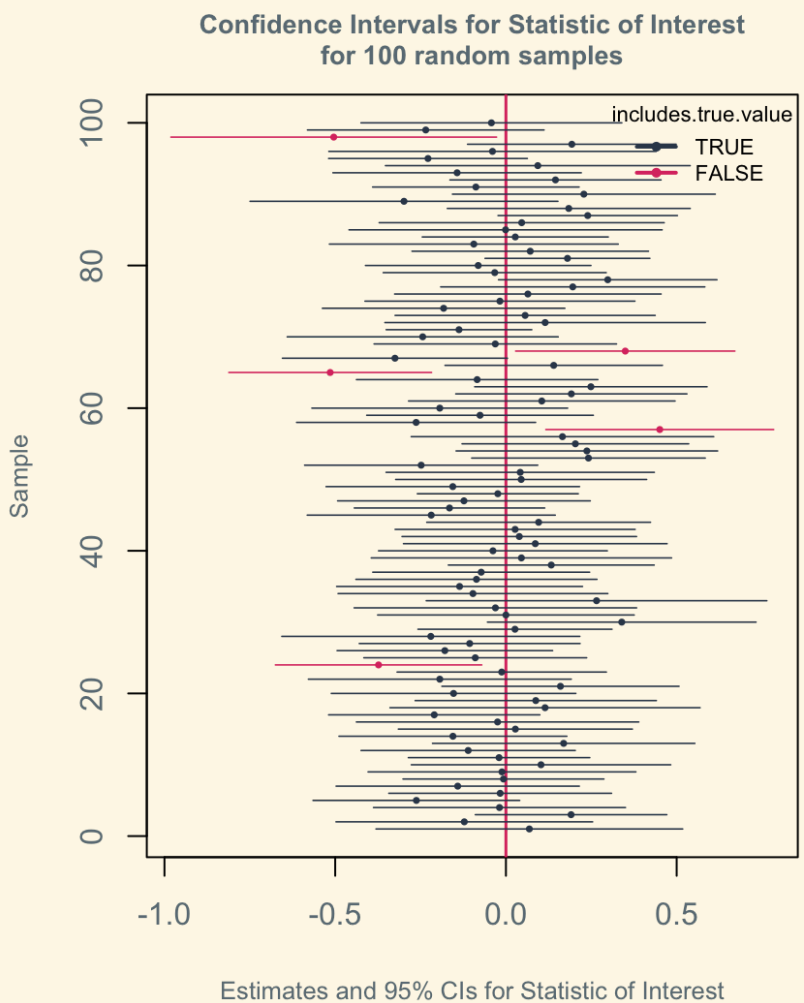
It's highly probable that μ is between 9.6 and 10.4. However, this statement **makes no probabilistic sense!** The unknown μ is **not** a random variable – it has a single fixed value, whether or not it is known to the statistician. Thus, μ is either in the interval or it's not.

Interpreting Confidence Intervals

The “trick” that has been devised is to move from a probabilistic statement about the unknown (but with a fixed value) **parameter**, to a probabilistic statement about the **method** of constructing the interval.

We say something like “the interval (9.6, 10.4) was constructed with a method that is 95% successful” – ie, if we repeated this method many times, 95% of those confidence intervals would cover the true value

This is (partly) where “frequentism” gets its name



Hypothesis Tests

A point **null hypothesis** is the statement that θ equals a specific value θ_0

$$H_0 : \theta = \theta_0 \qquad \text{vs} \qquad H_a : \theta \neq \theta_0$$

The null represents the skeptic's position — often a default assumption of "no effect" — we require the data to provide evidence against it

Many nulls are natural: the context tells you what θ_0 should be

- A/B test: the difference in means is zero ($\mu_A - \mu_B = 0$)
- Regression coefficient: a predictor has no effect ($\beta_k = 0$)
- Proportion: a coin is fair ($p = 0.5$)

A **hypothesis test** is a decision rule that uses a test statistic to either reject or fail to reject H_0

Test Statistics

A **test statistic** $T = T(Y_1, \ldots, Y_n)$ is a function of the data, typically chosen so that it

- it's small when H_0 is true
- the larger it is, the stronger the evidence against H_0

The null hypothesis is rejected if $T(\boldsymbol{Y}) > c$ for some **critical value** c.
Else you fail to reject

In practice, you specify α (the significance level), which determines c

The most common test statistic is the **z-stat** (or *t-stat*):

$$z = \frac{\hat{\theta} - \theta_0}{SE(\hat{\theta})}$$

P-Values

Let $t_{obs} = T(\boldsymbol{y})$ for the observed data $\boldsymbol{y}$

The **p-value** is the probability of observing a test statistic at least as extreme as t_{obs}, assuming the null hypothesis is true:

$$p = P\big(T(\boldsymbol{Y}) \geq t_{obs} \mid H_0\big)$$

It can be interpreted as the minimal significance level for which you would reject the null for a given value of $T(\boldsymbol{Y})$

A small p-value means: "data this extreme would be very unlikely if H_0 were true"

It does **not** mean: "the probability that H_0 is true is small" — H_0 is either true or it isn't, just like θ is either in the CI or it isn't

Statistical Significance

An estimate is said to be **statistically significantly** different from the point null hypothesis if

- The null hypothesis is rejected (ie $t_{obs} > c$)
- The p-value is less than α
- The confidence interval does not contain the point null hypothesized value θ_0

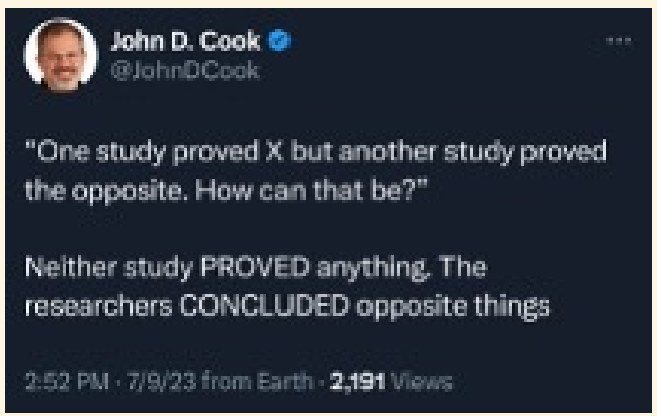
Advice

Simply reporting the binary statement “(not) statistically significant” and adding asterisks to tables is **terrible practice**. Instead:

- **Better to report p-values**
- **Better still to report standard errors and confidence intervals**

That said, business decisions are binary — launch or don’t, change or don’t. When you act on a CI, you’re implicitly performing a hypothesis test. CIs are better for communicating results; tests are unavoidable when deciding

One “statistically significant” result **does not *prove* anything**. It could be a Type I error. Results from one analysis should be combined with other results. Results must be interpreted in the context of the analyses.



Common Pitfalls

Statistical significance $\neq$ practical significance With large n, even trivially small effects become "significant." Always ask: *is the effect large enough to matter?*

"Fail to reject" $\neq$ "accept H_0" Absence of evidence is not evidence of absence — you may simply lack the sample size to detect a real effect

Multiple comparisons If you test 20 hypotheses at $\alpha = 0.05$, you expect one false positive even when all nulls are true. Running many tests and reporting only the significant ones is called "p-hacking" (also sometimes "lying")

Confusing SE with SD The SD describes the spread of the *data*; the SE describes the precision of an *estimate*. Often related by $SE \approx SD/\sqrt{n}$, but they answer different questions

Recap

- We focus on the conditional distribution $f_{Y|\boldsymbol{X}}$ rather than the full joint
- Assume rows are iid and $f_{Y|\boldsymbol{X}}$ belongs to a parametric family $\mathcal{F}_\theta$
- The statistical task becomes: learn θ from the data
- An estimator $\hat{\theta}$ is a function of the random sample; its distribution is the sampling distribution
- The bootstrap lets us estimate the SE of $\hat{\theta}$ by resampling
- By the LLN, $\hat{\theta}$ is consistent; by the CLT, its sampling distribution is approximately normal
- A confidence interval $\hat{\theta} \pm 1.96 \cdot SE$ covers the true θ 95% of the time
- A hypothesis test compares $\hat{\theta}$ to a null value θ_0 via a test statistic
- The p-value measures how surprising our data are if H_0 is true